

Why NeuroShift Matters

Need analysis with real population data + full analysis of base paper (Chamberland et al., IEEE EMBC 2024) | July 2026

About 100 million people live with stroke effects worldwide. India has millions with locomotor disability and rising stroke prevalence. The base paper uses EMG + vision to fix wrong gestures for prostheses. NeuroShift uses EMG + vision to decide which home device to control - or abstain - for safer independent living.

Part A - Population Need (Real Data)

A1. Problem

Many people cannot easily use switches, remotes, or touchscreens due to stroke-related weakness/paralysis, spinal cord injury, limb amputation, age-related decline, or neuromuscular conditions. NeuroShift helps them control home devices using remaining muscle signals (EMG) + camera context, with safe abstention when unsure.

A2. Global numbers

Metric	Number	Source
People living with stroke effects	~93.8M (2021); ~104.8M (2023 est.)	WHO / GBD / WSO
New strokes per year	~11.9 to 13.2 million	WHO / GBD
Stroke deaths per year	~6.8 to 7.3 million	WHO / GBD
Lifetime stroke risk	1 in 4 adults over age 25	WSO / WHO
Global stroke economic cost	>US\$890B / year (~0.66% GDP)	WSO / Lancet Neurology
People with spinal cord injury	~20.6 million (2019)	GBD / Lancet Neurology
Traumatic limb amputation survivors	~57.7 million (2017)	GBD-based study

- Stroke is a leading cause of long-term motor disability.
- WHO: survivors often have persistent motor deficits and loss of independence.
- Indian rehab survey analysis: nearly 90% of stroke survivors report movement impairment.
- Literature often cites 15-30% permanently disabled after stroke.

A3. India-specific numbers

Metric	Number	Source
Persons with disability	26.8 million	Census of India 2011
Movement / locomotor disability share	~20.3% of PwD (largest category)	Census 2011
Disability prevalence	~2.2% of population	NSS 76th Round (2018)
India new stroke cases (2021)	~1.25 million (~10% of global)	GBD / Lancet reporting
India stroke prevalence	~4.4M (1990) to ~9.4M (2021) (+47%)	GBD / Lancet reporting
India stroke incidence range	~108-172 per 100,000 / year	Jones et al., Int J Stroke 2022

- Most global stroke burden is in low- and middle-income countries.
- Care is often family-based; caregiver anxiety/depression reported in 17-50% in Indian reviews.
- Low-cost stack (MyoWare + webcam + ESP32) fits Indian college/startup constraints.

A4. Who NeuroShift helps

Group	Problem with normal controls	How NeuroShift helps
Stroke survivors	Hard to use remotes/switches	EMG effort + scene context
Partial paralysis /	One-sided motor loss	Remaining muscle as confirmation
Spinal cord injury (incomplete)	Limited voluntary movement	Low-effort volition + affordances
Upper-limb amputees	Related to base paper domain	Extend grips idea to home devices
Older adults / weak hand mobility	Fatigue, tremor, complex gestures	No big gesture dictionary
Caregivers / families	High ADL assistance burden	Safer automation + decision logs

A5. Usefulness in all ways

Dimension	Why useful
Social / humanitarian	Supports independent living for motor-disabled populations
Clinical / rehab	Extends EMG+vision from prosthesis literature to home ADLs
Research / IEEE	Clear novelty: home intention + Act/Abstain + FAR metric
Economic	Stroke costs hundreds of billions; reducing caregiver load has value
India relevance	Rising stroke prevalence + large locomotor disability population
Startup / portfolio	Problem -> prototype -> paper -> MVP path

Part B - Base Paper Analysis

Title: Visual Scene Understanding for Enhanced EMG Gesture Recognition
Authors: Chamberland, Labbe, Tam, Scheme, Gosselin | Venue: IEEE EMBC 2024
DOI: 10.1109/EMBC53108.2024.10782354

B1. What they do (plain words)

- HD-EMG predicts 6 hand gestures using SDCNN.
- Head-mounted webcam + YOLOv8 detects everyday objects.
- MediaPipe tracks the hand.
- Object Likelihood Filtering (OLF) decides likely target object.
- Vision limits allowed grips to those compatible with that object.
- Result: fewer false gesture detections; user stays in control.
- Domain: myoelectric PROSTHESIS control (not smart home).

B2. Technical stack

Module	Their choice
EMG sensor	EMaGer 64-channel HD-EMG @ 1 kHz
EMG model	SDCNN (Siamese CNN) + centroid classifier
Gestures	open, tripod, power, thumbs-up, pinch, pointed index
Camera	EMEET C960 head-mounted, 640x480, ~10 fps
Vision	Custom YOLOv8-small, 28 object classes
Hand tracking	MediaPipe Hands
Fusion	OLF enables object-related grips for EMG selection

B3. Results

SDCNN intra-session accuracy: 98.6%. YOLO mAP@[,5,.95]: 0.368.

Pilot: 1 able-bodied subject, 2 scenarios x 20 trials.

Scenario	Transition errors fixed	Static errors fixed
#1 mug -> apple	100% (5.7 -> 0)	68.2% (10.7 -> 3.4)
#2 bottle -> phone	60.7% (5.6 -> 2.2)	93.0% (7.1 -> 0.5)

Takeaway: vision context strongly reduces false EMG gesture predictions.

B4. Strengths

- Clear multimodal architecture (EMG + YOLO + hand tracking).
- Preserves user agency (EMG primary; vision curates options).
- Real-time proof with measurable error reduction.
- Confidence-aware EMG model when gestures are ambiguous.

B5. Limitations = NeuroShift starting point

Base paper limitation	NeuroShift response
Predicts gestures, not home actions	Predict appliance intention (lamp/fan/plug)
Prosthesis grip selection domain	Assistive smart-home independent living
64-channel HD-EMG (expensive)	Sparse MyoWare (low-cost, realistic)
Vision filters allowed grips	Vision proposes device affordances
No Act/Abstain for IoT safety	Selective abstention + False Actuation Rate
Pilot = 1 able-bodied subject	Multi-user NeuroShift-ADL pilot
Hard with many nearby objects	Multi-object ambiguity is our core test
No habit prior / caregiver audit	Sem 8 temporal prior + explanations

Authors themselves note challenges: crowded scenes, occlusion, unknown objects, and better EMG/vision confidence fusion.

B6. Extension statement (use in reports)

Chamberland et al. (EMBC 2024) showed that visual scene understanding can reduce false EMG gesture detections for prosthesis control by enabling only object-compatible grips. NeuroShift extends this multimodal paradigm from gesture validation for prostheses to home-device intention inference for assistive living. Using sparse EMG volition, ambient object/pose affordances, and personal priors with selective abstention, NeuroShift targets lower false smart-home actuations for people with motor disabilities.

Part C - 90-second explanation for guide

- Need: ~100M live with stroke effects globally; India ~9.4M stroke prevalence and millions with locomotor disability.
- Base paper: EMG + camera reduces wrong GESTURES for prostheses.
- Gap: that does not solve wrong HOME DEVICE actuations.
- Our solution: camera proposes feasible devices; EMG confirms effort; Act or Abstain.
- Why useful: social impact + research novelty + India-feasible hardware + IEEE path.

Bottom line

NeuroShift is useful because the population need is large and growing. The base paper proves EMG + vision works for safer gesture control. Our contribution turns that idea into safer home intention control for independent living.